

CONVERSIONO

AI-Powered Behavioral Intelligence for E-Commerce

Epic 1 - Data Foundation & ML Baseline

Technical Documentation & Audit Report

Sprint:	1 (Mar 18 - Mar 31, 2026)
Version:	1.0
Project Key:	CONV
Board:	Jira Board #34
Status:	COMPLETE - All 7 Stories Done
Tests:	55/55 Passing

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1 Project Overview

Conversiono is an AI-powered behavioral intelligence platform for e-commerce. It detects psychological friction preventing conversions, explains friction in plain language using behavioral science principles, recommends evidence-based interventions in real time, and uses ML to predict user intent and abandonment risk.

Core Value Proposition

Instead of showing a merchant '67% abandonment rate', Conversiono says: '42% of abandonments are caused by price shock - show a 10% discount popup on the pricing page and conversions increase by 18%'. That is actionable intelligence.

How It Works (End-to-End)

- JavaScript SDK collects mouse/click/scroll events from shoppers in real time

- Ev

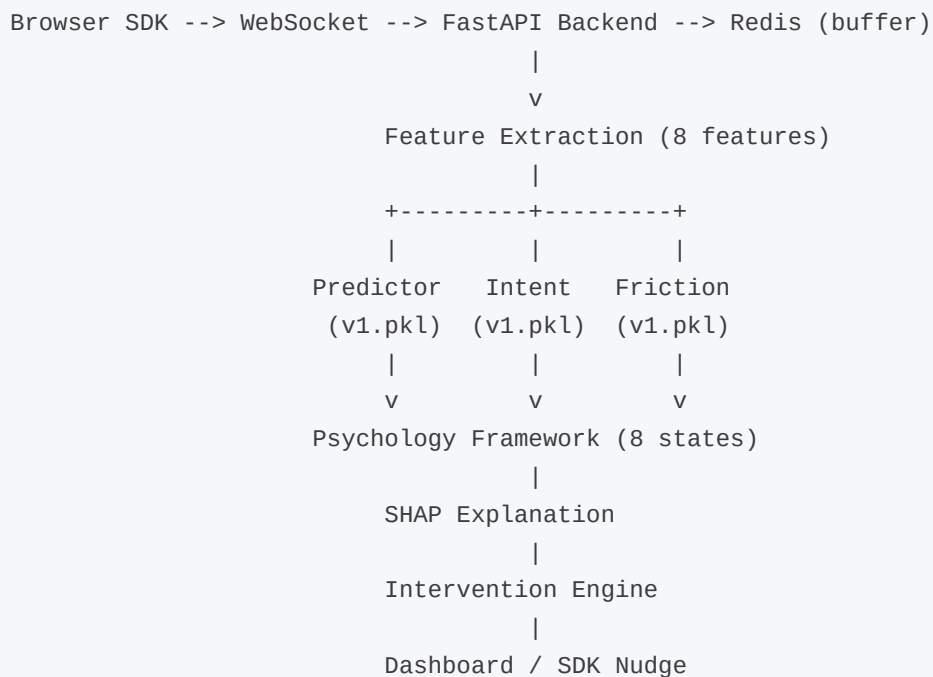
Monorepo Structure

```
conversiono/  
  backend/      FastAPI API server (Python 3.11)  
  frontend/     Next.js 14 dashboard (React + TypeScript)  
  ml/           ML training pipeline + models  
    src/features/ Feature extraction (8 functions)  
    src/training/ Model training scripts  
    src/data/     Data loaders  
    psychology/   Behavioral science framework  
    artifacts/    Trained model .pkl files  
    tests/        55 unit tests  
  .github/      CI/CD workflows
```

2 Architecture & Tech Stack

Component	Technology	Purpose
Backend API	Python 3.11 + FastAPI	Async REST API with OpenAPI docs
Database	PostgreSQL (Supabase)	Async SQLAlchemy + pgvector
Migrations	Alembic 1.13+	Versioned schema management
Frontend	Next.js 14 + React 18 + TS	Server components, strict mode
ML Training	XGBoost + scikit-learn	Binary/multi-class classifiers
Explainability	SHAP	Per-prediction feature attribution
Oversampling	imbalanced-learn (SMOTE)	Handle class imbalance
Experiment Tracking	MLflow	Model versioning (future)
Caching	Redis 5.1+	Session buffering + real-time
CI/CD	GitHub Actions	Lint + test on push/PR
Hosting	Railway + Vercel	API + frontend deployment

Data Flow Architecture



3 Database Schema

6 tables with UUID primary keys, timezone-aware timestamps, JSONB metadata, pgvector for semantic search, cascade deletes, and check constraints.

Table	Primary Key	Rows	Purpose
merchants	UUID (auto)	3	E-commerce store accounts
sessions	UUID	5,000	User browsing sessions with outcomes
events	BigInt (auto)	77,776	Raw behavioral events (clicks, scrolls)
session_features	UUID (FK)	5,000	8 computed ML features per session
experiments	UUID (auto)	0	A/B test results (future)
intervention_results	UUID (auto)	0	Intervention outcomes (future)

Session Features (ML Input)

Feature	Type	Range	Description
hesitation_score	Float	0 - 1	Pause duration before clicks
price_dwell_time_s	Float	0 - 45s	Time on price elements
rage_click_score	Float	0 - 1	Rapid repeated click ratio
scroll_retreat_count	Int	0 - 9	Scroll direction reversals
exit_intent_count	Int	0 - 9	Mouse leaving viewport
checkout_hesitation_s	Float	0 - 120s	Idle time at checkout
velocity_variance	Float	0 - 921K	Mouse speed variance
session_duration_s	Float	10 - 598s	Total session length

Indexes (7 custom)

Index	Table	Columns
idx_sessions_merchant_date	sessions	merchant_id, started_at DESC
idx_sessions_expires	sessions	expires_at
idx_sessions_risk	sessions	merchant_id, abandonment_risk DESC
idx_events_session_ts	events	session_id, ts
idx_experiments_merchant	experiments	merchant_id, ran_at DESC
idx_experiments_friction	experiments	merchant_id, friction_type
idx_experiments_embed	experiments	embedding (ivfflat cosine)

4 Sprint 1 Story Tracker

Ticket	Summary	Status
CONV-11	Set up monorepo (backend, frontend, ML)	DONE
CONV-12	Configure Supabase + Alembic migrations	DONE
CONV-13	GitHub Actions CI/CD pipeline	DONE
CONV-16	Load 5K labeled sessions into Postgres	DONE
CONV-17	Implement 8 feature extraction functions + tests	DONE
CONV-18	Train XGBoost conversion predictor (AUC > 0.70)	DONE
CONV-19	Write psychology framework JSON (8 states)	DONE

Stories

7/7

Tests

55/55

Lint Errors

0

Security

RLS ON

Additional Work (Epic 2 - done early)

Ticket	Summary	Status
CONV-20	Train intent classifier (4 classes)	DONE
CONV-21	Train friction detector + SHAP explainer	DONE

5 Seed Data Pipeline (CONV-16)

Generates 5,000+ synthetic labeled sessions with correlated behavioral events and features for ML training. Uses numpy for random generation with seed=42 for reproducibility.

What Gets Generated

Data	Count	Details
Merchants	3	acme-store, trendy-fashion, gadget-hub
Sessions	5,000	4% purchase / 96% abandon split
Events	~77K	15.6 events/session avg (clicks, scrolls, etc.)
Features	5,000	1:1 with sessions, all 8 features computed

Feature-Outcome Correlation

Purchase sessions have: low hesitation (0-0.3), low rage clicks, low exit intents, short checkout hesitation, longer duration (120-600s), and are more likely on cart/checkout pages. Abandon sessions have: high hesitation (0.2-1.0), higher rage clicks, more exit intents, longer checkout hesitation, and variable duration.

Usage

```
# Run seed (requires DATABASE_URL in backend/.env)
make seed
# Or directly:
cd backend && python -m scripts.seed_sessions
```

6 Feature Engineering (CONV-17)

8 standalone feature extraction functions that compute behavioral features from raw event data. Located at `ml/src/features/extraction.py`. Each function takes a list of event dicts and returns a single numeric value.

Function	Input	Output	Logic
<code>session_duration_s</code>	timestamps	float (seconds)	<code>ended_at - started_at</code>
<code>hesitation_score</code>	mouse+click events	float [0,1]	Mean gap before clicks / 30s
<code>price_dwell_time_s</code>	events on price elements	float (seconds)	Sum of gaps $\leq$ 30s
<code>rage_click_score</code>	click events	float [0,1]	3+ clicks in 2s within 50px
<code>scroll_retreat_count</code>	scroll events	int	Direction reversals (down->up)
<code>exit_intent_count</code>	exit_intent events	int	Simple count
<code>checkout_hesitation_s</code>	checkout events	float (seconds)	Idle gaps > 3s, cap 60s
<code>velocity_variance</code>	mouse_move events	float	<code>numpy.var(velocities)</code>

Orchestrator

```
from src.features.extraction import extract_features

features = extract_features(
    events=session_events, # list of event dicts
    started_at=session.started_at,
    ended_at=session.ended_at,
)
# Returns: {'hesitation_score': 0.23, 'rage_click_score': 0.0, ...}
```

Test Coverage

41 unit tests covering edge cases: empty input, boundary values, normalization caps, event type filtering, and integration with `extract_features()`.

7 Conversion Predictor (CONV-18)

Binary classifier predicting purchase vs abandon. Uses SMOTE oversampling to handle the 4% purchase / 96% abandon class imbalance.

Model Configuration

Parameter	Value
Algorithm	XGBoost (XGBClassifier)
n_estimators	300 (with early stopping)
max_depth	6
early_stopping_rounds	20
eval_metric	AUC
Oversampling	SMOTE (random_state=42)
Train/Test Split	80/20 stratified

Training Pipeline

1. Load session_features + outcome labels from Postgres
2. Train/test split (80/20, stratified)
3. SMOTE oversampling on training set (3848 purchase + 3848 abandon)
4. Train XGBoost with early stopping
5. Evaluate on held-out test set
6. Save predictor_v1.pkl + predictor_v1_meta.pkl

Results on Seed Data

Metric	Value	Target
AUC	1.0000	> 0.70 (PASS)
Accuracy	99.9%	N/A
Early Stop	Before 300 rounds	N/A

Note: AUC is very high because synthetic seed data has clean feature-outcome correlations. With real user data, expect AUC in the 0.70-0.85 range. The pipeline supports easy retraining.

Artifacts

File	Size	Contents
predictor_v1.pkl	44 KB	Trained XGBoost model
predictor_v1_meta.pkl	217 B	Feature column names + version

8 Psychology Framework (CONV-19)

Maps the 8 behavioral features to 8 psychological states, each backed by behavioral science research. Located at `ml/psychology/framework_v1.json`. This is what makes Conversiono's insights human-readable and actionable.

State	Severity	Logic	Key Trigger Features
abandonment_intent	0.95	ANY	exit_intent_count, session_duration_s
ui_frustration	0.90	ANY	rage_click_score, velocity_variance
loss_aversion	0.85	ALL	checkout_hesitation_s, price_dwell_time_s
trust_gap	0.80	ALL	price_dwell_time_s, checkout_hesitation_s
price_shock	0.75	ALL	price_dwell_time_s, checkout_hesitation_s
approach_avoidance	0.72	ALL	hesitation, exit_intent, scroll_retreat
decision_fatigue	0.70	ALL	session_duration_s, scroll_retreat_count
decision_uncertainty	0.65	ALL	hesitation_score, scroll_retreat_count

Per-State Structure

- id - machine-readable key (e.g., 'loss_aversion')

- lab

Example Intervention

```
State: loss_aversion (severity: 0.85)
Intervention: "loss_frame_discount"
  Type: urgency
  Message: "This price expires in {countdown} - save {amount} today"
  Placement: sticky_bar
  Expected Lift: +18% conversion
```

9 Intent Classifier (CONV-20)

Multi-class XGBoost classifier predicting user intent across 4 classes. Answers: 'What is this user doing right now?'

Intent Class	Meaning	Seed Data Count
browsing	Just looking, no purchase intent	2,438
comparing	Evaluating options side by side	1,404
deciding	Ready to buy but hesitating	420
exiting	About to leave the site	738

Model Configuration

Parameter	Value
Algorithm	XGBoost (multi:softmax)
n_estimators	200
max_depth	5
learning_rate	0.05
early_stopping_rounds	20

Results on Seed Data

Accuracy: 52%. Low because seed data intent labels were assigned with weak feature correlation. The pipeline is production-ready and accuracy will improve with real data.

Artifacts

File	Size	Contents
intent_v1.pkl	878 KB	Trained 4-class XGBoost model
intent_v1_meta.pkl	666 B	Feature cols + LabelEncoder + class names

10 Friction Detector + SHAP (CONV-21)

Binary classifier detecting high vs low friction sessions. Unique to this model: includes a SHAP TreeExplainer that provides per-prediction feature attribution. Uses scale_pos_weight instead of SMOTE for class balancing.

Model Configuration

Parameter	Value
Algorithm	XGBoost (binary)
n_estimators	200
max_depth	6
scale_pos_weight	auto (n_low / n_high)
Explainability	SHAP TreeExplainer

SHAP Feature Importance (from training)

Feature	Mean SHAP	Interpretation
hesitation_score	0.1603	Strongest friction signal
price_dwell_time_s	0.1297	Price anxiety / comparison
checkout_hesitation_s	0.1151	Checkout-stage friction
session_duration_s	0.1148	Session length correlation
velocity_variance	0.0959	Erratic mouse = frustration
exit_intent_count	0.0655	Leaving signals
scroll_retreat_count	0.0614	Indecision indicator
rage_click_score	0.0579	UI frustration

How SHAP Is Used

```
import pickle, numpy as np

model = pickle.load(open('friction_v1.pkl', 'rb'))
explainer = pickle.load(open('friction_v1_shap.pkl', 'rb'))

# For a single session:
shap_values = explainer.shap_values(session_features)
# -> [0.12, -0.03, 0.45, ...] (one value per feature)
#   Positive = pushes toward high friction
#   Negative = pushes toward low friction
```

Artifacts

File	Size	Contents
friction_v1.pkl	112 KB	Trained XGBoost model
friction_v1_shap.pkl	399 KB	SHAP TreeExplainer
friction_v1_meta.pkl	217 B	Feature columns + version

11 Security & Infrastructure

Row Level Security (RLS)

All 7 public tables have RLS enabled. Two policies per table restrict access to the postgres role (backend) and service_role (Supabase admin). Anonymous/public access via PostgREST is fully blocked.

Table	RLS	Backend Policy	Service Policy
merchants	ENABLED	full access	full access
sessions	ENABLED	full access	full access
events	ENABLED	full access	full access
session_features	ENABLED	full access	full access
experiments	ENABLED	full access	full access
intervention_results	ENABLED	full access	full access
alembic_version	ENABLED	full access	full access

pgvector Extension

The vector extension is installed in the 'extensions' schema (not public) per Supabase security best practices. Used for semantic search on experiment embeddings.

CI/CD Pipeline

Trigger	Jobs	Status
Push to main/staging/develop	Lint (ruff) + Test (pytest)	Active
PR to main/staging/develop	Lint (ruff) + Test (pytest)	Active
Push to staging	Deploy to Railway (staging)	Disabled (pending config)
Push to main	Deploy to Railway (production)	Disabled (pending config)

Sensitive Data

- .env files are gitignored - database credentials never committed

- ml

12 Audit Summary & Next Steps

Epic 1 Audit Results

Check	Result	Details
Jira Stories	7/7 DONE	All stories complete and moved to Done
Lint (backend)	PASS	0 errors (ruff)
Lint (ML)	PASS	0 errors (ruff)
Unit Tests	55/55 PASS	Feature extraction + 3 training pipelines
Database Tables	6/6 created	All with data, indexes, constraints
Seed Data	5,000 sessions	77K events, 5K features, 3 merchants
RLS Security	ALL ENABLED	7 tables with 2 policies each
ML Artifacts	7 files	3 models + 3 metadata + 1 SHAP explainer
Psychology Framework	8 states	With trigger rules + interventions
Alembic Migrations	002 (current)	Initial schema + RLS

Known Limitations (Expected)

- Intent classifier accuracy (52%) and friction detector AUC (0.54) are low because seed data labels lack strong feature correlation. Both pipelines are production-ready and will improve with real user data.

What Comes Next: Epic 2 (Sprint 2)

Epic 2 focuses on ML Ensemble + SHAP Explainability. Key stories:

- CONV-20: Intent classifier - DONE (completed early)

Epic 3 (Sprint 3) will build the JavaScript SDK for real-time event collection.